Why is *The Origin of Species* such a great book? First of all, because it convincingly demonstrates the fact of evolution: it provides a vast and well-chosen body of evidence showing that existing animals and plants cannot have been separately created in their present forms, but must have evolved from earlier forms by slow transformation. And secondly, because the theory of natural selection, which the *Origin* so fully and so lucidly expounds, provides a mechanism by which such transformation could and would automatically be produced. Natural selection rendered evolution scientifically intelligible: it was this more than anything else which convinced professional biologists like Sir Joseph Hooker, T. H. Huxley and Ernst Haeckel.

In neither case did Darwin shrink from drawing the most general conclusions. To begin with, he realized that evolution must be a universal phenomenon. If different species of groundfinches or armadillos could be produced by evolution from a common ancestor, then, given enough time, the same must hold for different families, orders, and classes and for the diversity of life as a whole: all living organisms must be related through their common descent from some simple original stock. Further, since all organisms vary, and all reproduce themselves in greater numbers than can survive, there must always be competition between variants; in other words, the principle of natural selection, too, is universally applicable.

For these reasons, Wallace himself rightly styled Darwin “the Newton of Natural History,” or, as we should now say, of Biology. Each introduced the ideas of unity, order, and universally applicable principle into an enormous realm of experience.

In 1859, the area of biological ignorance was very large. Nothing was known of the mechanisms of fertilization, heredity and variation, nor of embryonic differentiation; the scientific study of animal behaviour, biogeography and ecology had scarcely begun; no good paleontological series, like those of horse or elephant evolution, had been discovered, nor any fossils bearing on the ancestry of man; and the time-scale admitted by geologists and physicists was grossly inadequate. Yet in spite of this, Darwin in the *Origin* gave a remarkably good general picture of the evolutionary process, and followed out the implications of natural selection in a quite astonishing way.

Thus he deduced that natural selection must inevitably bring about the “improvement” of organisms, improvement which, as he characteristically added, was always in relation to the conditions of life. This, though he did not himself claim it, was in fact another universal biological law, covering